

5(a)(i)	amplitude of the carrier wave varies	M1
	in synchrony with the displacement of the (information) signal	A1
5(a)(ii)	Any 2 from: • fewer transmitters needed / each transmitter can cover a greater distance • more stations can share waveband • transmitters and receivers are cheaper	B2

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Question	Answer	Marks
5(b)(i)	$\lambda = \frac{v}{f}$ $= \frac{3.0 \times 10^8}{1.5 \times 10^6} = 200 \text{ m}$	A1
5(b)(ii)	10 kHz	B1
5(c)	1520 kHz	B1

Question	Answer	Marks
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